
ECSE 4660/6660: Internetworking of Things

Lecture 12: Link Layer
Fundamentals – Part IV: BLE,
ZigBee

Ish Jain
ECSE, RPI

Slides Acknowledgement [Pannuto, Ghena, Campbell] from UCSD

Spring 2026

Administrative

- Use Piazza for any questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Lab
 - **TA hours 4-5 PM** daily in 5312 lab
 - Remote TA hours temporarily
 - Lab 3 due **Fri, 2/27 (BLE, Zbee)**
- HW2 on **LMS** **due Mon 2/23**
 - No Late submission allowed
- Remote class **2/26, 3/23**

- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Next Steps
 - **Work with TA to get the hardware needed (respond on piazza)**
 - Mid-term report due 3/20 (4 weeks)
 - In-class presentation+2min video 4/20, 4/23
 - Live-demo 4/27
 - Final code/report 5/1

Spectrally Agile Cellular Wireless Systems

Sundeep Rangan

Professor, Electrical & Computer
Engineering
NYU Tandon

Mercer Distinguished Lecture Series

LOW 3051

Wed, February 25, 2026 at 4:00 PM

The spectrum landscape for future cellular wireless systems is emerging as a patchwork of bands spanning wide ranges of frequencies including, most recently, new frequency bands in the so-called upper mid-band. Traditional exclusive licenses are increasingly difficult to find as these bands must be shared among competing services, most notably rapidly expanding satellite mega-constellations, and vital defense and public safety RADAR and tactical communications systems. These developments motivate novel spectrally agile systems for spectrum sensing, band selection, inter-network coordination, and resiliency to hostile attacks. In this talk, we will review recent results on developing spectrally agile systems in these bands including (1) capacity analyses with adaptive frequency hopping; (2) algorithms for satellite and terrestrial co-existence; (3) RF and digital architectures for reconfigurable, wideband massive MIMO platforms; and (4) experimental platforms for FR3.

Prof. Dr. Sundeep Rangan received the [B.A.Sc.](#) at the University of Waterloo, Canada and the M.Sc. and Ph.D. at the University of California, Berkeley, all in Electrical Engineering. He has held postdoctoral appointments at the University of Michigan, Ann Arbor and Bell Labs. In 2000, he co-founded (with four others) Flarion Technologies, a spin off of Bell Labs, that developed Flash OFDM, one of the first cellular OFDM data systems and pre-cursor to 4G systems including LTE and WiMAX. In 2006, Flarion was acquired by Qualcomm Technologies where Dr. Rangan was a Senior Director of Engineering involved in OFDM infrastructure products. He joined the ECE department at NYU Tandon (formerly NYU Polytechnic) in 2010. He is a Fellow of the IEEE and Associate Director of NYU WIRELESS, an academic-industry research center researching next-generation wireless systems.



References

1. Bluetooth Low Energy (BLE)

- Reference

N. Hunn, “**Essentials of Short-Range Wireless**,” Cambridge University Press, 2010.

Chapter 7: BLE

2. IEEE 802.15.4 and Zigbee

- Reference

N. Hunn, “**Essentials of Short-Range Wireless**,” Cambridge University Press, 2010.

Chapter 6: IEEE 802.15.4 and ZigBee

Bluetooth Low Energy (BLE)

Bluetooth Low Energy Resources

- Good walkthrough of BLE:
 - <https://www.silabs.com/documents/public/user-guides/ug103-14-fundamentals-ble.pdf>
- [[5.2 specification](#)] [[4.2 specification](#)] (link to PDF download)
 - Also: [[Supplement v9](#)]
- More boots-on-the-ground view:
 - <https://inst.eecs.berkeley.edu/~ee290c/sp18/lec/Lecture7A.pdf>
 - From a team that has implemented BLE HW several times
 - https://download.ni.com/evaluation/rf/intro_to_bluetooth_test.pdf

Outline

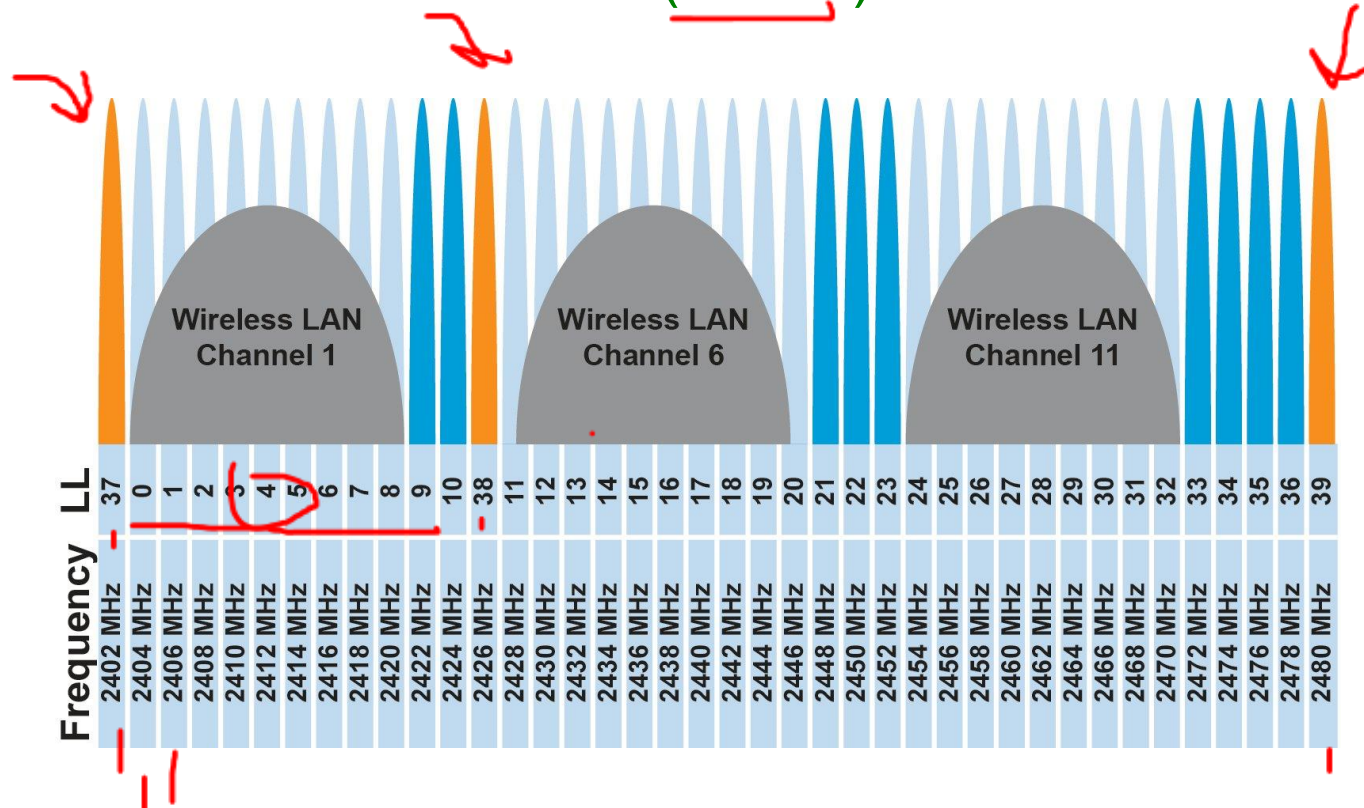
- BLE Background
- **BLE Layers**
 - Physical Layer
 - Link Layer
- BLE roles
 - Advertising
 - Scanning

BLE frequency usage and channels

- 2.4 GHz carrier, forty 2-MHz channels, 1 Mbps data rate

→ 37, 38, 39 for advertising

- 0-36 for connection (FHSS)

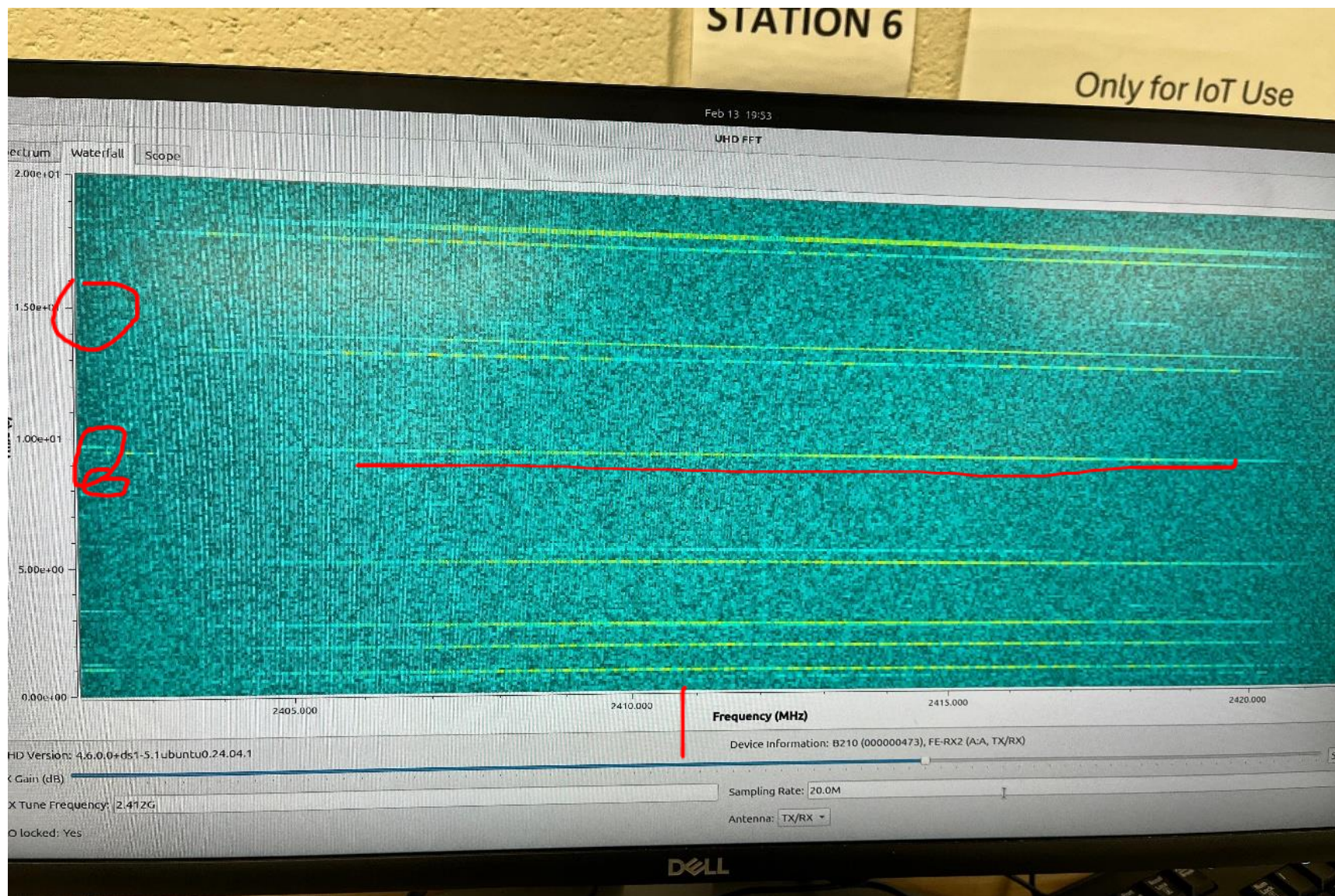


Why doesn't BLE avoid WiFi altogether?

Can't on 2.4 GHz

40

Lab 3: Spectrum analysis for BLE with USRP setup



Outline

- BLE Background
- BLE Layers
 - Physical Layer
 - Link Layer
- **BLE roles**
 - Advertising 
 - Scanning 

Advertising: BLE's discovery mechanism

- Make nearby devices aware of advertiser's existence
- Communicate some information from or about advertiser
- Traditional purpose is to enable connections, but this is also useful for general communication

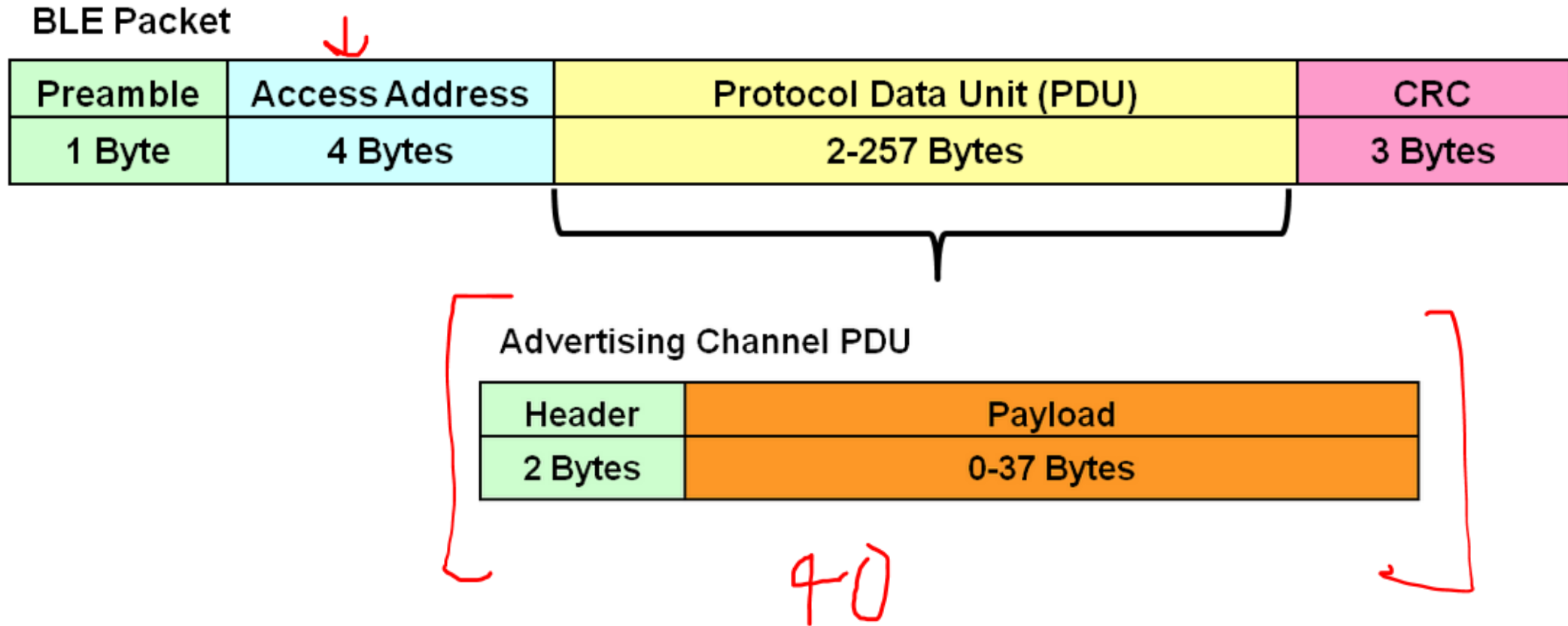
■ Advertisements

- Periodic broadcast messages with data

■ Scan Requests/Responses

- Scanner sends responses after getting a request
 - Only occurs when scanner is listening
- Almost literally “bonus advertisement data”

Advertising packet layering



BLE advertising header

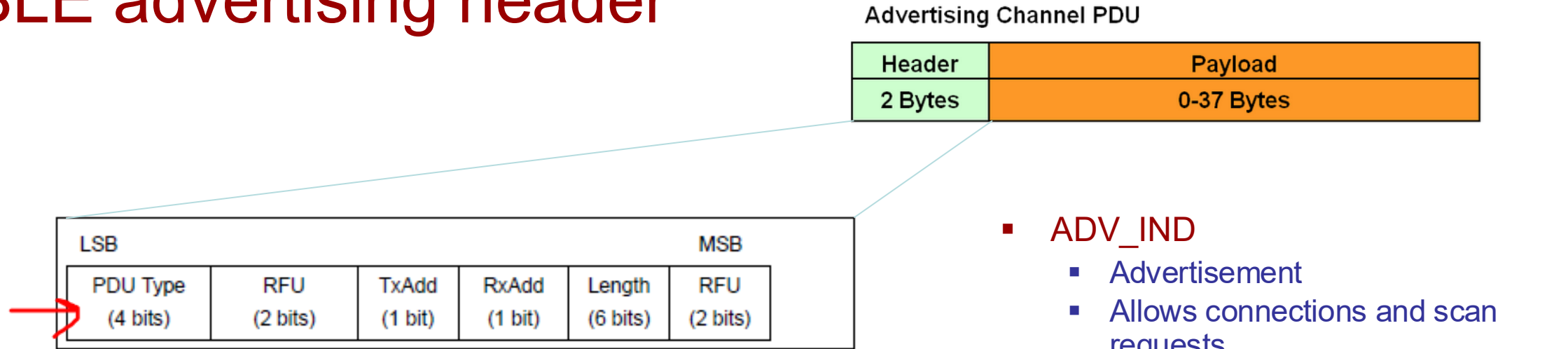


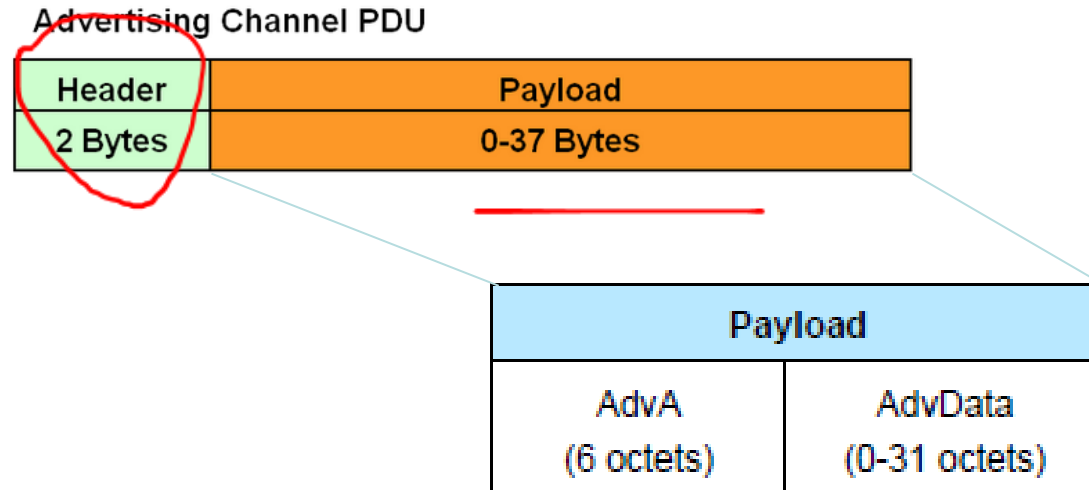
Figure 2.3: Advertising channel PDU Header

PDU Type $b_3b_2b_1b_0$	Packet Name
0000	ADV_IND
0001	ADV_DIRECT_IND
0010	ADV_NONCONN_IND
0011	SCAN_REQ
0100	SCAN_RSP
0101	CONNECT_REQ
0110	ADV_SCAN_IND
0111-1111	Reserved

Table 2.1: Advertising channel PDU Header's PDU Type field encoding

- **ADV_IND**
 - Advertisement
 - Allows connections and scan requests
- **ADV_NONCONN_IND**
 - Advertisement
 - No connections or scan requests
- **ADV_SCAN_IND**
 - Advertisement
 - No connections but allows scan requests
- **SCAN_REQ**
 - Scan request
- **SCAN_RSP**
 - Scan response

Advertisement payloads



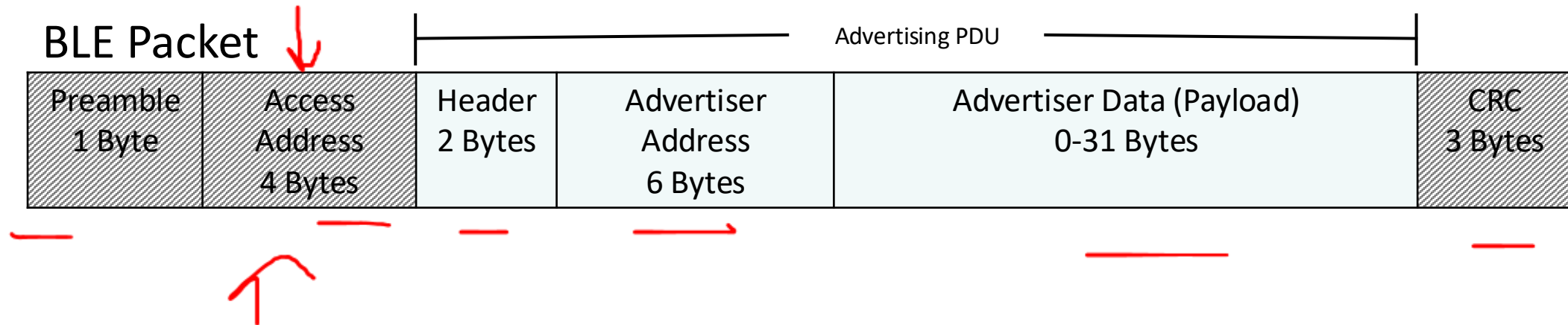
- AdvA – address of the advertiser
 - TxAdd bit from header specifies if this is a “public” or “random” address
- Remaining up to 31 bytes are available for use

Advertiser addresses

- Public and private address forms
- Public
 - 48 bits: 24-bits of company ID, 24-bits of company assigned number
 - Literally the same MAC address scheme as Ethernet and WiFi
- Private
 - Top two MSbs specify type
 - 46 bits of random
 - 46 bits of hash of an identity key
- Why have the two types?
 - Privacy

Advertisement Packet

- Putting it all together, up to 47 bytes total:



Scan Requests and Responses

- Scan request
 - Just the two addresses: the scanner's and the advertiser's

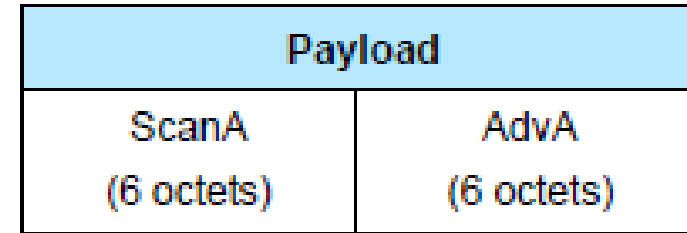


Figure 2.8: SCAN_REQ PDU Payload

- Scan response
 - Identical to an advertisement
 - But only occurs after a request

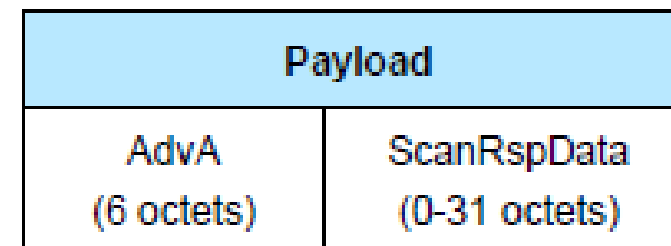
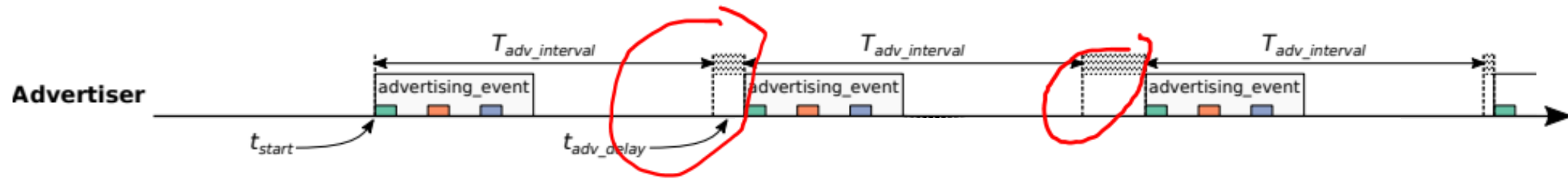


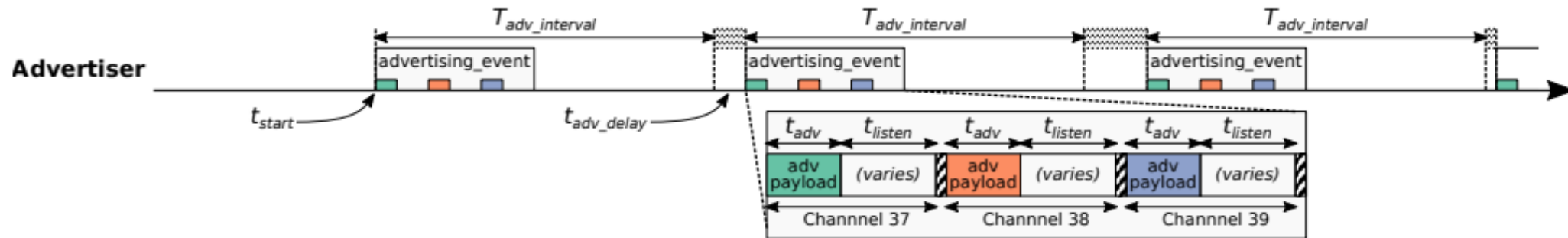
Figure 2.9: SCAN_RSP PDU payload

Advertising timing



- Advertising Events occur periodically [20ms – 10.24 s] (or longer)
 - User picks the rate as a tradeoff of energy and discovery latency
- Plus a random delay after each instance [0-10ms]
 - **Why?**
 - Avoid repeat collisions.

Advertising event



- Three transmissions, one on each advertising channel
 - Always in the same order
- Transmission, followed by listening window on that same channel
 - Requests will be sent ≥ 150 us (Inter-Frame Spacing, IFS) after Tx
 - Followed by a retune to the next channel frequency
- This short listen window is the magic “low energy” part



Preserving energy in communication

- Most energy is spent listening
 - This is due primarily to how long listening durations are compared to transmissions
- Question: advertising vs. listening in BLE:
 - How much more energy is used listening for 1 second versus transmitting 30 bytes three times?
 - Radio uses 60 mW when active (TX or RX)
 - 1 Mbps data rate

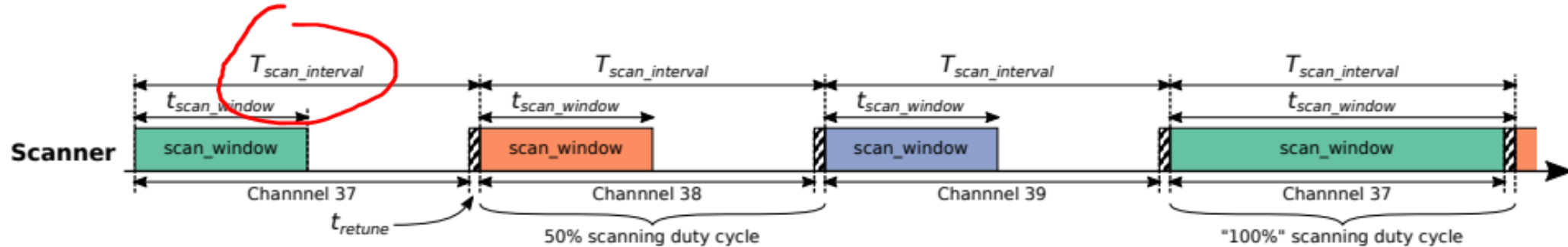
Preserving energy in communication

- 30 bytes = 240 bits
- 240 bits / 1 Mbps = 0.240 ms
- 0.24 * 3 = 0.72 ms (three channels)
- 1 s / 0.720 ms = 1389
- 1389x!!

Outline

- BLE Background
- BLE Layers
 - Physical Layer
 - Link Layer
- BLE roles
 - Advertising
 - **Scanning**

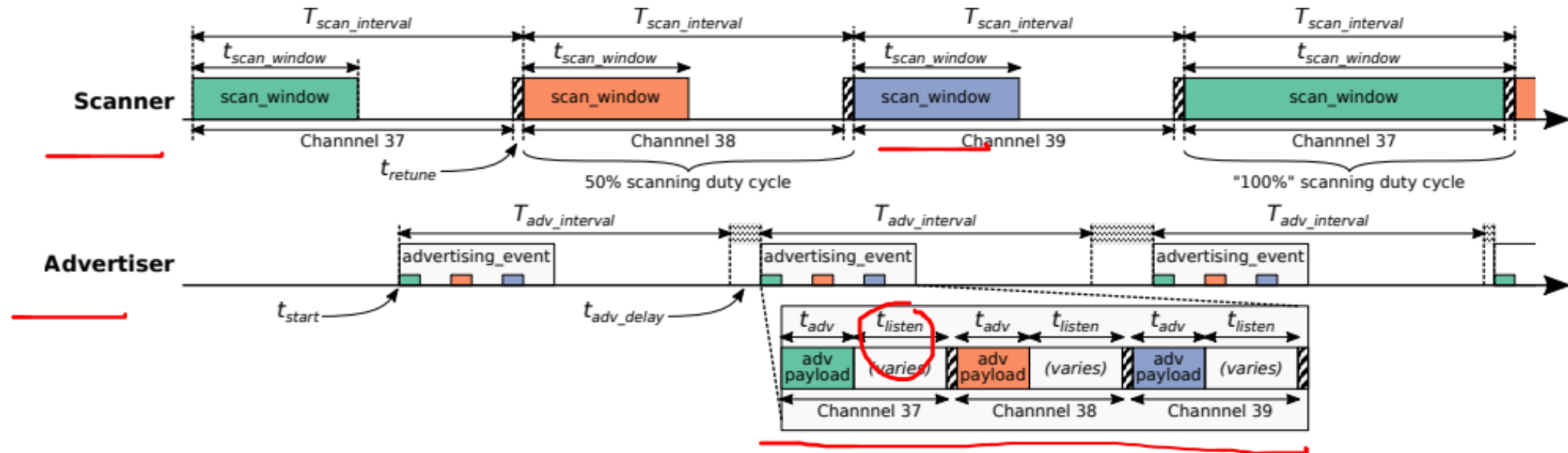
Scanning Pattern



- Iterate through channels, listening for advertisements
 - $T_{\text{scan_interval}}$ controls rate at which channels are changed
 - $T_{\text{scan_window}}$ controls duty cycle of listening
- Why listen at a low duty cycle?
 - Save energy

Putting it all together

- Advertisements are received when the channel of the scan window and the channel of the advertisement overlap in time (and space)

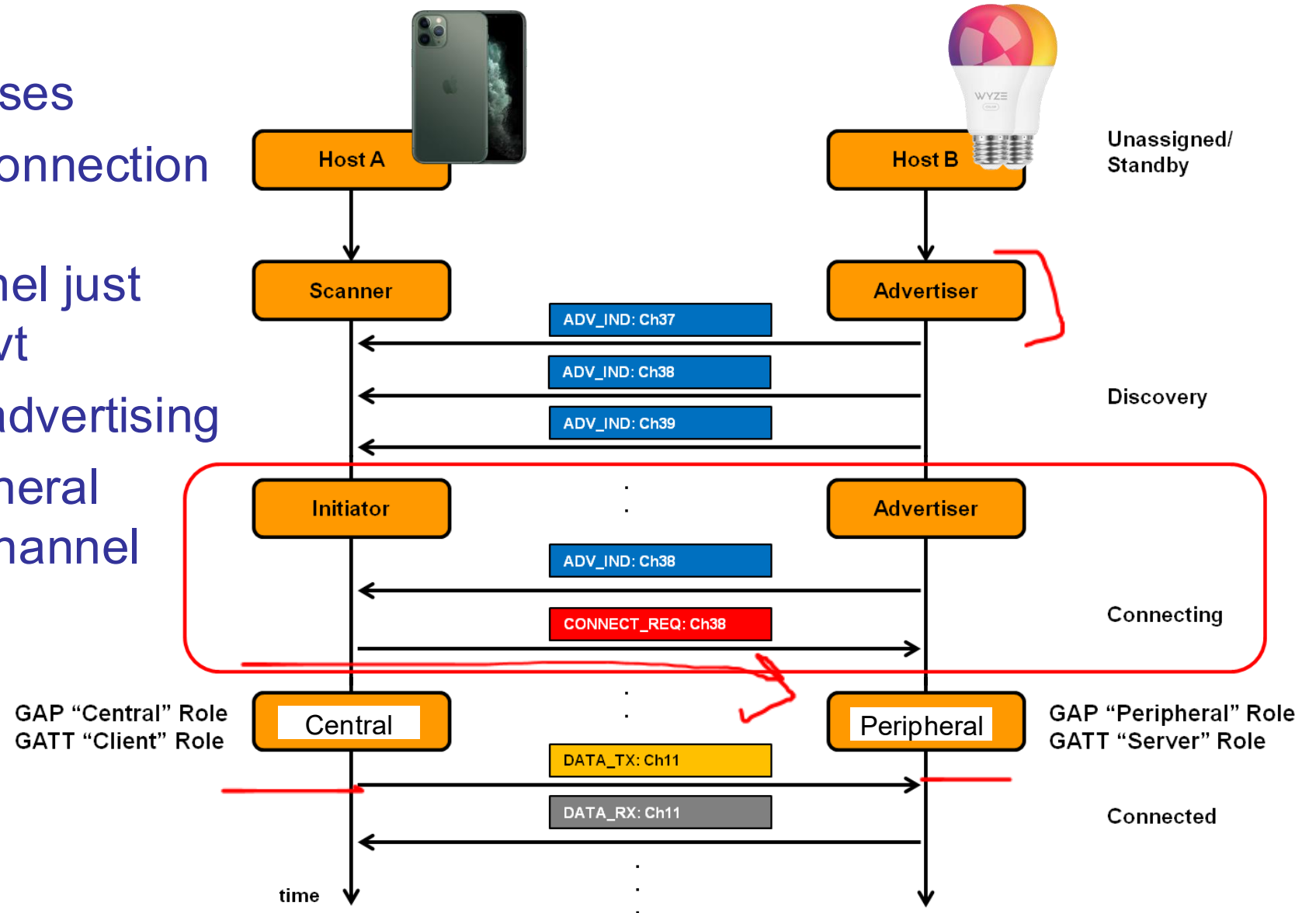


A warning about scanning expectations

- Scanners will NOT receive 100% of packets sent
 - Even ignoring range issues
- Packets are lost due to (in roughly descending order):
 - Duty cycle
 - Sharing 2.4 GHz antenna with WiFi
 - Retune period after each scanning interval
 - Dropped packets in the receive software
 - Packet collisions

Connection Timeline

- Peripheral advertises
- Central sends a connection request packet on advertising channel just after receiving advt
- Peripheral stops advertising
- Central and peripheral switch to a data channel



Lab 3: BLE packet analysis tool

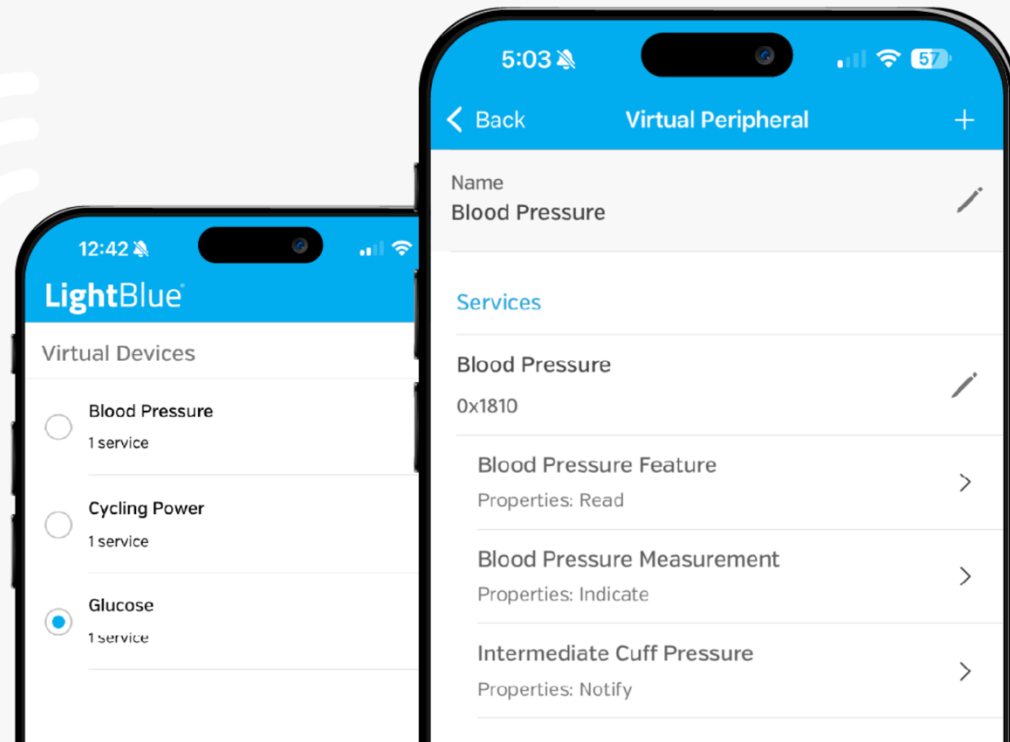


Your Ultimate BLE Debugging & Development Toolkit

For Mobile & Embedded Software Developers



Available on iOS, macOS, and Android.



Next topic: IEEE 802.15.4 and Zigbee

Goals

- Introduce the IEEE 802.15.4 standard
 - Specifically designed for low power applications
- Overview of physical layer details
- Exploration of link layer
 - Network topologies
 - Communication structure

References

- 802.15.4 Specification [[2006](#)]
 - “Part 15.4: Wireless Medium Access Control (MAC) and Physical Layer (PHY) Specifications for Low-Rate Wireless Personal Area Networks (WPANs)”

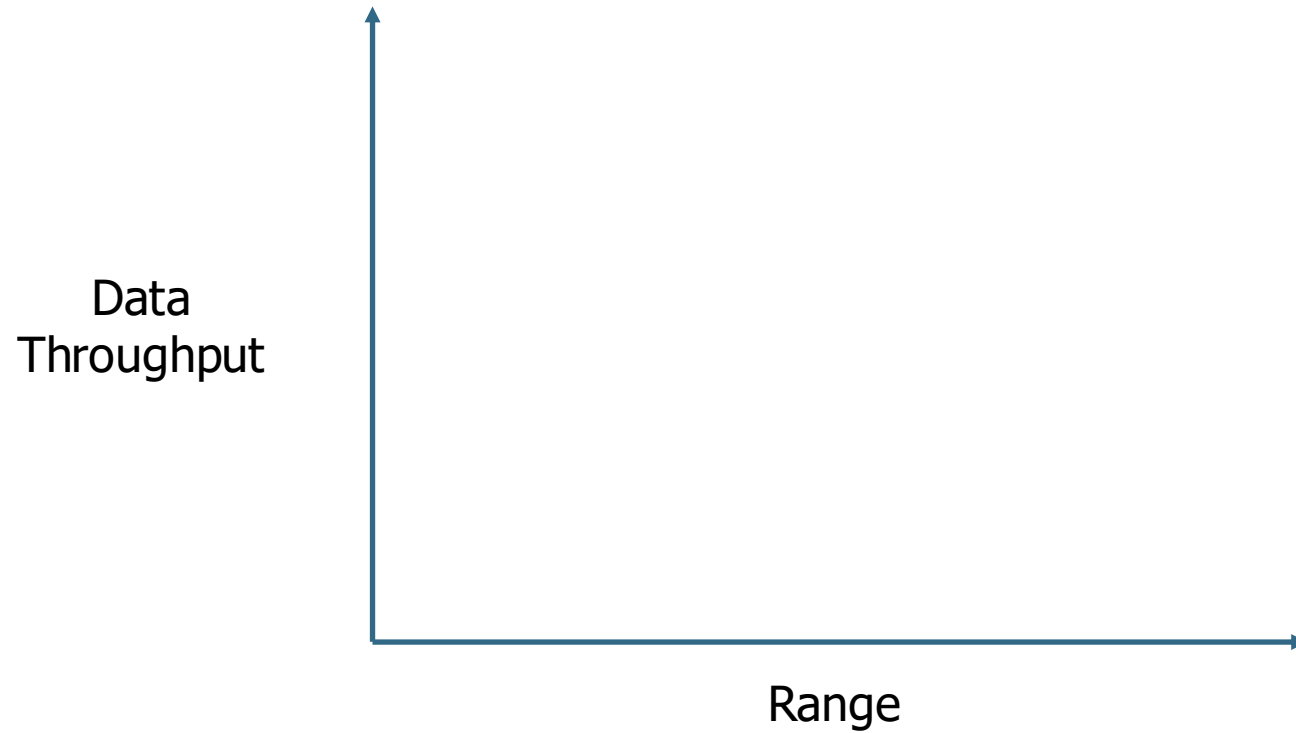
Other helpful references:

- [Paper introducing the 802.15.4 draft](#)
- [NXP 802.15.4 Stack User Guide](#)
- [2005 presentation on 802.15.4](#)

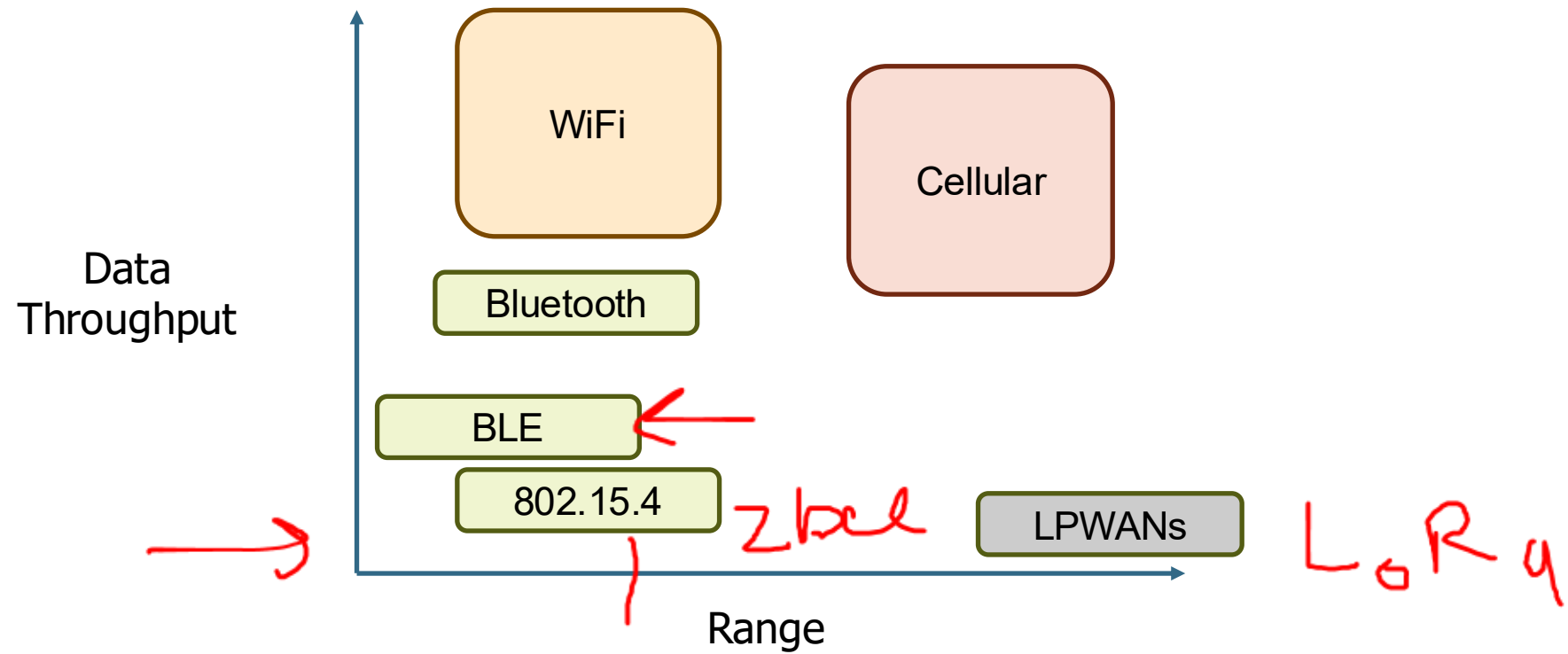
Outline

- **Overview**
- Physical Layer
- Link Layer

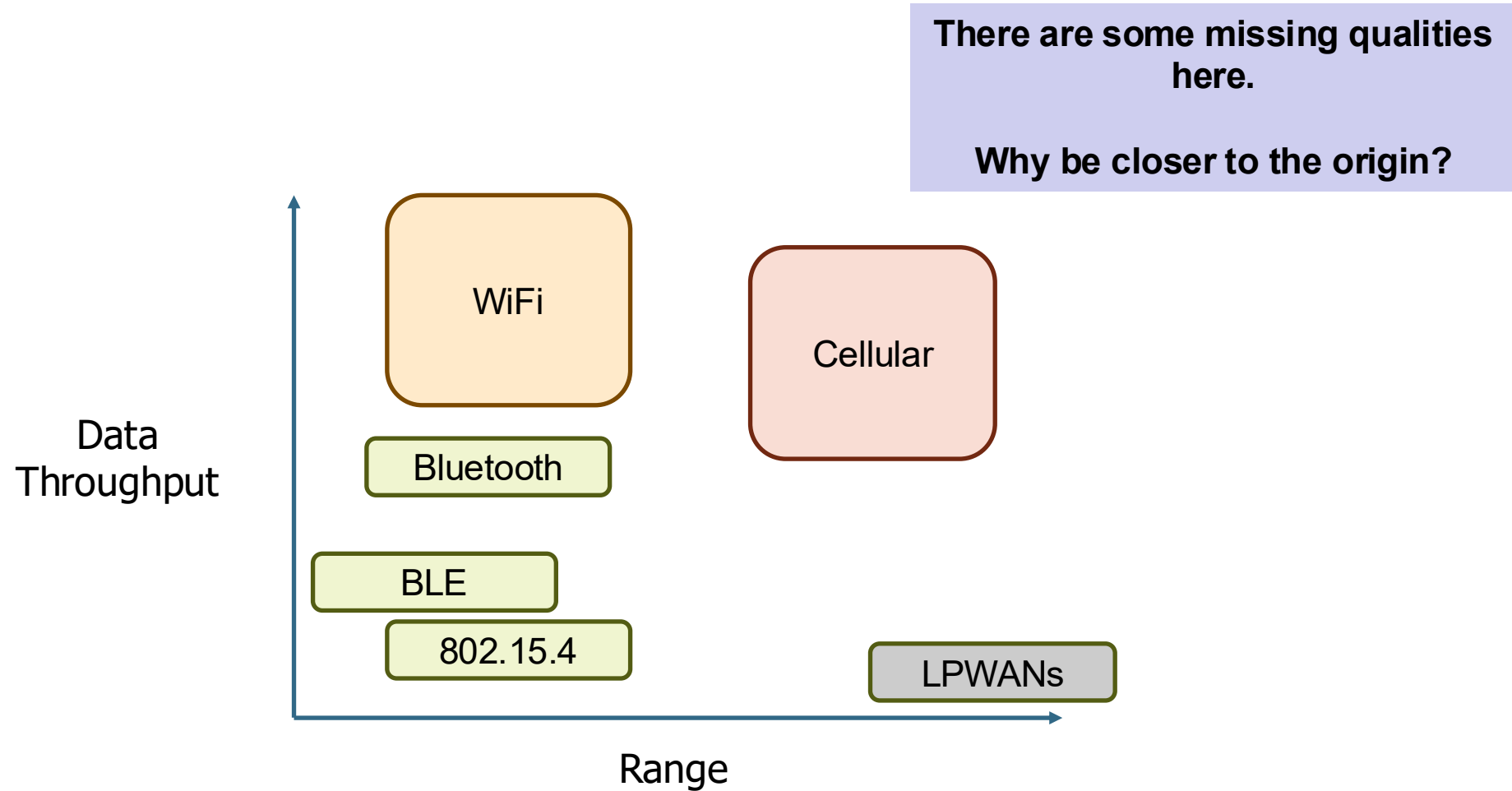
Comparison of networks



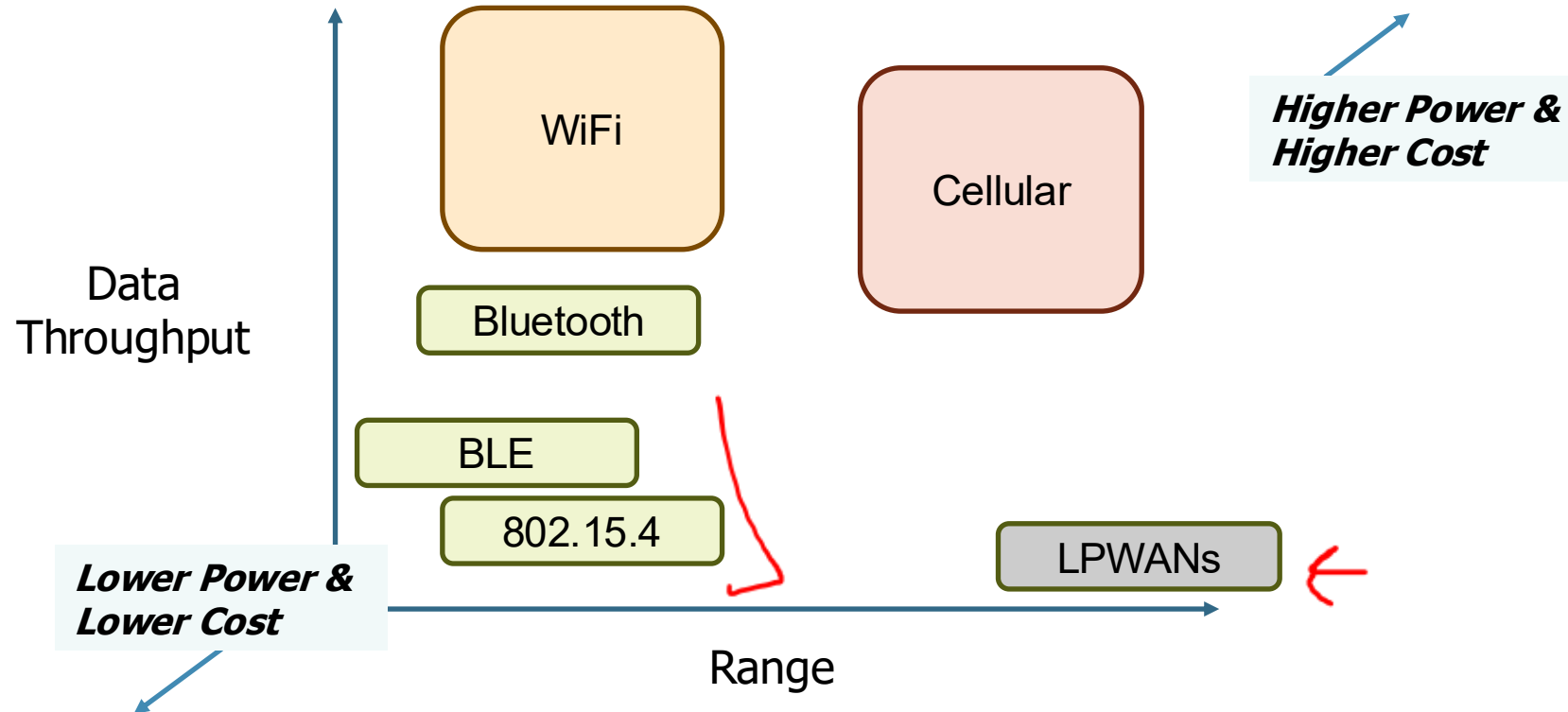
Comparison of networks



Comparison of networks



Comparison of networks



IEEE 802

- Network standards for variable-sized packets
 - Ethernet
 - WiFi
 - WPANs
- E.g. **not** networks that send periodic constant-sized packets
- Specifies PHY and Link layers
- Another example standard:
 - IEEE 754: Floating Point

Name	Description	Status
IEEE 802.1	Higher Layer LAN Protocols Working Group	Active
IEEE 802.2	LLC	Disbanded
IEEE 802.3	Ethernet	Active
IEEE 802.4	Token bus	Disbanded
IEEE 802.5	Token Ring MAC layer	Disbanded
IEEE 802.6	MANs (DQDB)	Disbanded
IEEE 802.7	Broadband LAN using Coaxial Cable	Disbanded
IEEE 802.8	Fiber Optic TAG	Disbanded
IEEE 802.9	Integrated Services LAN (ISLAN or isoEthernet)	Disbanded
IEEE 802.10	Interoperable LAN Security	Disbanded
IEEE 802.11	Wireless LAN (WLAN) & Mesh (Wi-Fi certification)	Active
IEEE 802.12	100BaseVG	Disbanded
IEEE 802.13	Unused ^[2]	reserved for Fast Ethernet development ^[3]
IEEE 802.14	Cable modems	Disbanded
IEEE 802.15	Wireless PAN	Active
IEEE 802.16	Broadband Wireless Access (WiMAX certification)	hibernating
IEEE 802.17	Resilient packet ring	Disbanded
IEEE 802.18	Radio Regulatory TAG	?
IEEE 802.19	Wireless Coexistence Working Group	?
IEEE 802.20	Mobile Broadband Wireless Access	Disbanded
IEEE 802.21	Media Independent Handoff	hibernating
IEEE 802.22	Wireless Regional Area Network	hibernating
IEEE 802.23	Emergency Services Working Group	Disbanded
IEEE 802.24	Vertical Applications TAG	?

← ZB

IEEE 802.15

- Wireless Personal-Area Networks (WPAN)
 - All the things within the workspace of a person
 - Conceptually smaller domain than the Local Area Network
 - Realistically about the same thing as a LAN
- Formerly included a Bluetooth spec
 - Bluetooth SIG took over governance

Name	Description	Status
IEEE 802.15.1	Bluetooth certification	Disbanded
IEEE 802.15.2	IEEE 802.15 and IEEE 802.11 coexistence	Hibernating ^[4]
IEEE 802.15.3	High-Rate wireless PAN (e.g., UWB, etc.)	?
IEEE 802.15.4	Low-Rate wireless PAN (e.g., <u>ZigBee</u> , <u>WirelessHART</u> , <u>MiWi</u> , etc.)	Active
IEEE 802.15.5	Mesh networking for WPAN	?
IEEE 802.15.6	Body area network	Active
IEEE 802.15.7	Visible light communications	?

802.15.4 (LR-WPANs) Overview “Low-Rate Wireless Personal Area Networks”

■ Goals

- “The IEEE 802.15 TG4 was chartered to investigate a low data rate solution with multi-month to multi-year battery life and very low complexity.” [TG4]

■ Applications

- “Potential applications are sensors, interactive toys, smart badges, remote controls, and home automation.” [TG4]
- Ultimately home automation, industrial control/monitoring, vehicular sensing, agriculture; really most machine-to-machine (M2M) sensor applications

■ Other contemporary technologies

- WiFi 802.11b and Bluetooth Classic
 - Too complex in specification and overachieving in capability

IEEE 802.15.4

- Low-Rate Wireless PAN
 - 250 kbps, ~100 m range
 - Radio hardware available with low-power and low-cost
- Specification: 2003
 - Also 2006, 2011, 2015, and 2020 revisions
 - Mostly various added capabilities such as extra PHY layers
 - Also define optional security, scheduling, and larger frame sizes
- We'll mostly work off of the 2006 version 
 - Thread is based on 2006 version
 - Zigbee is based on the original 2003 version
 - Roughly 200 pages of meaningful specification (100 of appendices)
 - Compare to 3000 pages of Bluetooth/BLE

Outline

- Overview
- **Physical Layer**
- Link Layer

802.15.4 Physical Layers

- Multiple options of physical layers are supported
 - We'll focus on 2.4 GHz (2400 MHz)

Table 1—Frequency bands and data rates

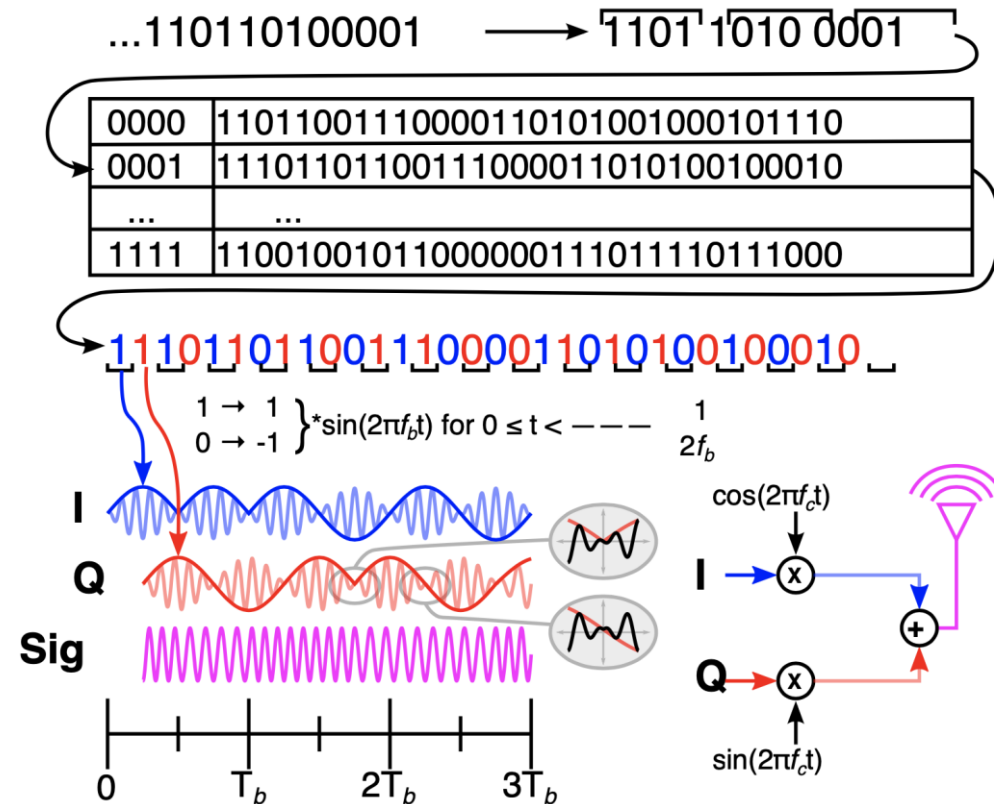
PHY (MHz)	Frequency band (MHz)	Spreading parameters		Data parameters		
		Chip rate (kchip/s)	Modulation	Bit rate (kb/s)	Symbol rate (ksymbol/s)	Symbols
868/915	868–868.6	300	BPSK	20	20	Binary
	902–928	600	BPSK	40	40	Binary
868/915 (optional)	868–868.6	400	ASK	250	12.5	20-bit PSSS
	902–928	1600	ASK	250	50	5-bit PSSS
868/915 (optional)	868–868.6	400	O-QPSK	100	25	16-ary Orthogonal
	902–928	1000	O-QPSK	250	62.5	16-ary Orthogonal
2450	2400–2483.5	2000	O-QPSK	250	62.5	16-ary Orthogonal

Physical Layer

PHY (MHz)	Frequency Band (MHz)	Spreading Parameters		Data Parameters		Channel Spacing (MHz)
		Chip Rate (kchips/s)	Modulation	Bit Rate (kb/s)	Symbol Rate (ksymbol/s)	
868/915	868–868.6 (Europe)	300/400/400	BPSK/ASK/O-QPSK	20/250/100	20/12.5/25	—
	902–928 (N. America)	600/1600/1000	BPSK/ASK/O-QPSK	40/250/25	40/50/62.5	2
2450 DSSS	2400–2483.5 (Global)	2000	O-QPSK	250	62.5	5

- O-QPSK modulation
 - Offset Quadrature Phase-Shift Keying
 - Twice the data rate of BPSK for same BER
 - Cost: most complicated design of receivers
 - Which is pretty minimal with all the transistors we've got
 - Plus the ability to reuse previous designs
 - 4 bits per symbol
- Symbols versus bits
 - A symbol is the unit of data transfer for a modulated signal
 - Does not necessarily correspond 1:1 with bits
 - The rate of symbols per second is a baudrate
- 802.15.4 bit rate at 2.4 GHz: 2000 kchips/s, which is 250 kbps, which is 62.5 kBaud

802.15.4 Modulation (@2.4 GHz f_c)



802.15.4 Modulation (@2.4 GHz f_c)

O-QPSK with half-sine shaping is MSK!

Input bit stream

...110110100001 → 1101 1010 0001

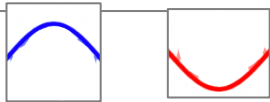
Broken into 4-bit *symbols*

Each bit of the PN code is called a *chip*

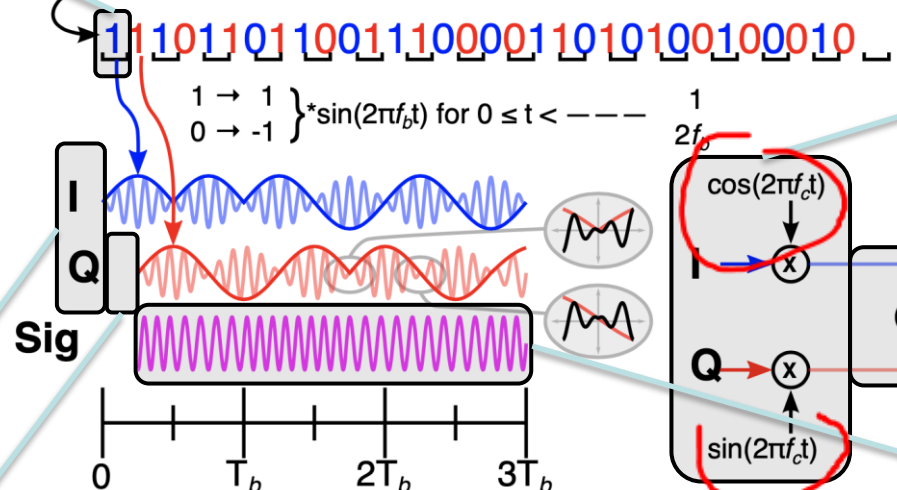
0000	11011001110000110101001000101110
0001	11101101100111000011010100100010
...	...
1111	11001001011000000111011110111000

Each *symbol* maps to a 32-bit *pseudo-noise code (PN-code)* or sometimes *pseudo-random sequence*

Each *chip* encodes half a sine wave



Chips alternate in-phase and quadrature



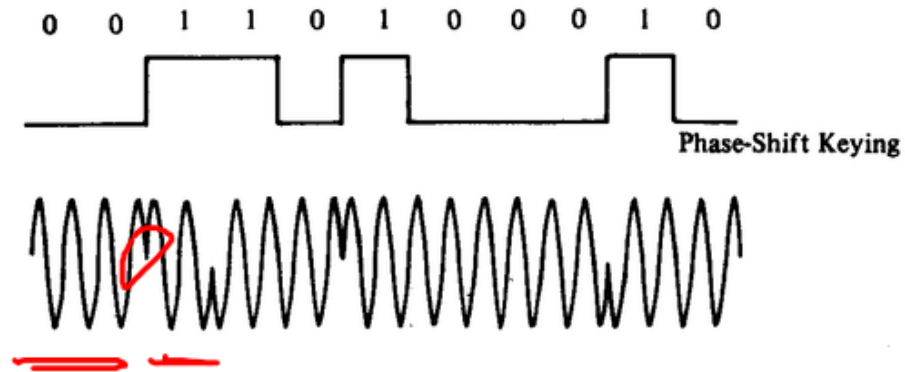
I and Q half-sines are *baseband*, which are mixed with the *carrier*

I and Q carriers are combined to create the final on-air *signal*

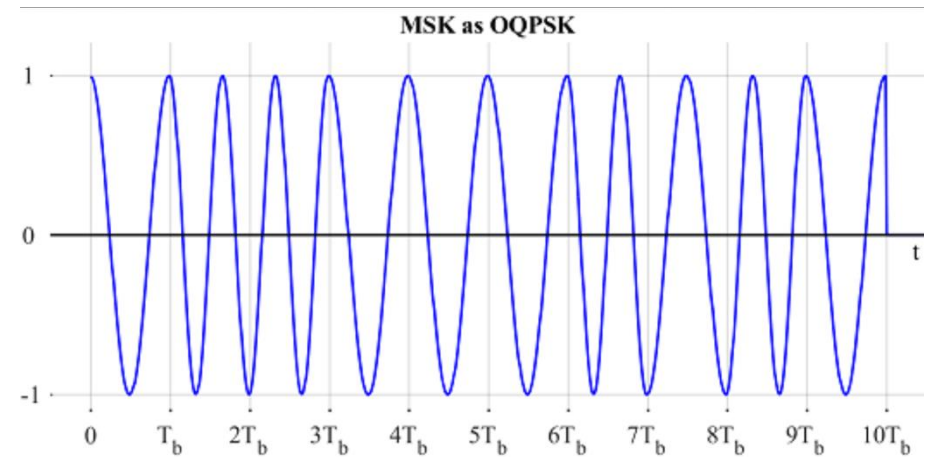
Signal is MSK, which is a special, optimal case of FSK!

Quadrature component is offset $\pi/2$

O-QPSK results in continuous wave



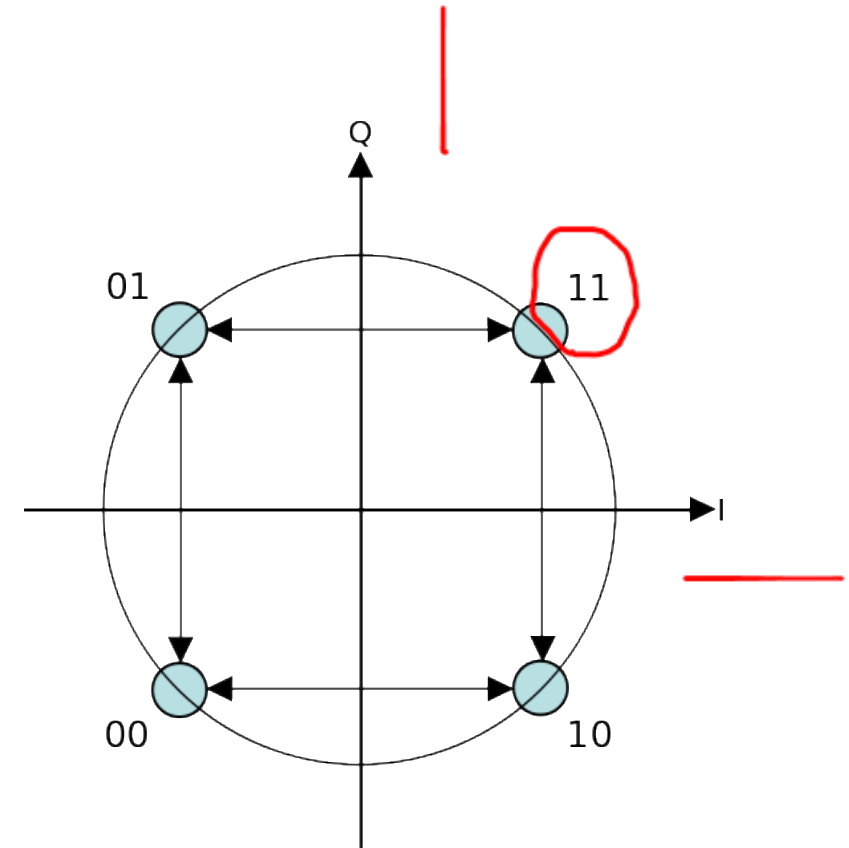
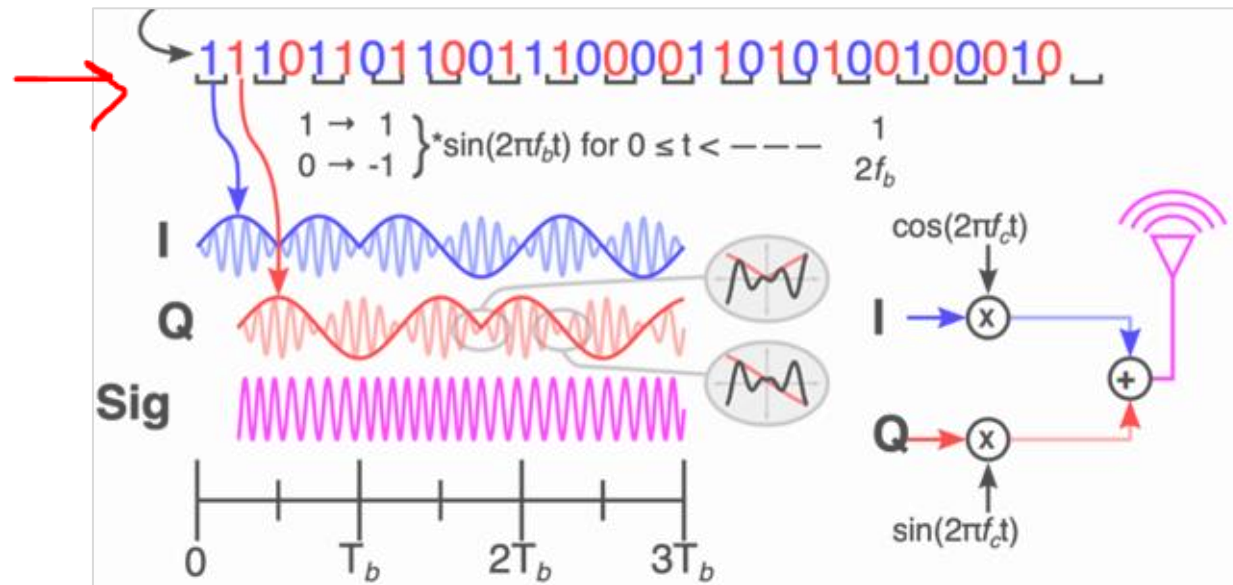
Standard BPSK



→ O-QPSK into Minimum Shift Keying (MSK)

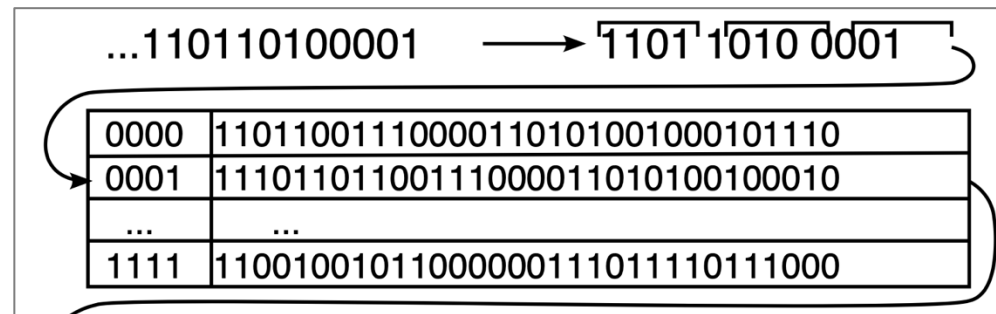
The magic of I and Q channels are that we get two dimensions

- This is called a “constellation diagram”



Why do we map symbols to chips?

- We took the 4 bits we want to send...
... and sent 32 bits instead??
- Why?

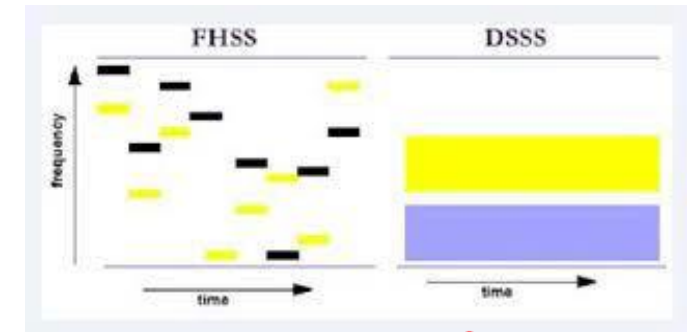


Direct Sequence Spread Spectrum (DSSS)

- Increases the signal bandwidth of a transmission beyond information bandwidth
 - Send sequences of chips, which are a translation of one symbol to a pattern of many bits
 - Chips are transmitted much faster than symbols, essentially increasing the data rate
- Enables better interference avoidance
 - Received bits are correlated against codes to see which is most likely
 - 802.15.4 tolerates 13-15 bit flips (almost half!)

Table 1. Zigbee symbol to chip mapping.

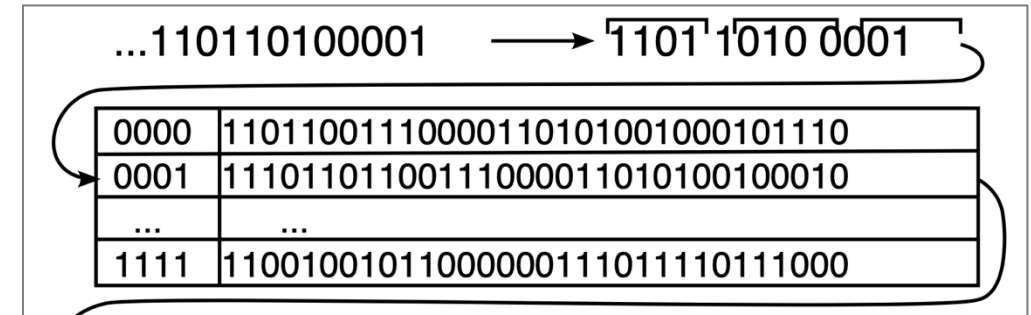
Zigbee Symbol	Chip Values ($c_0c_1\dots c_{30}c_{31}$)
0000	11011001110000110101001000101110
1000	11101101100111000011010100100010
0100	00101110110110011100000110101010
1100	00100010111011011001110000110101
0010	01010010001011101101100111000011
1010	00110101001000101110110110011100
0110	11000011010100100010111011011001
1110	10011000011010100100010111011101



BLE 2MHz

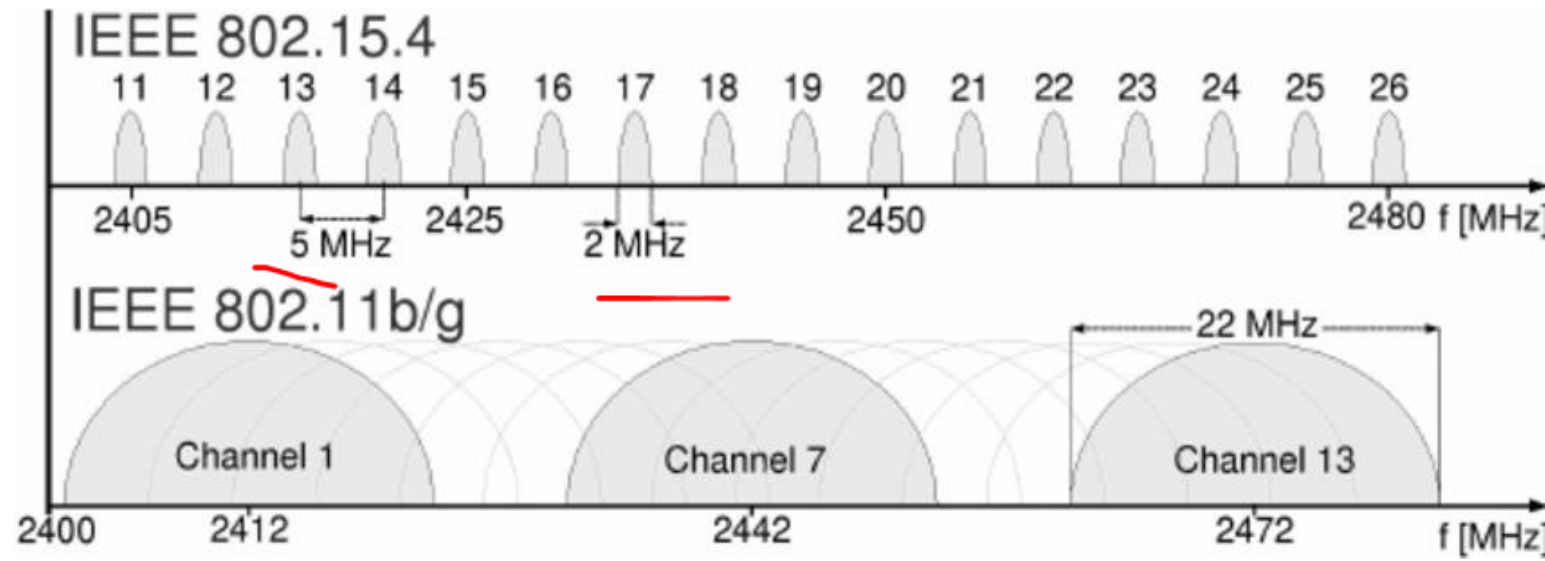
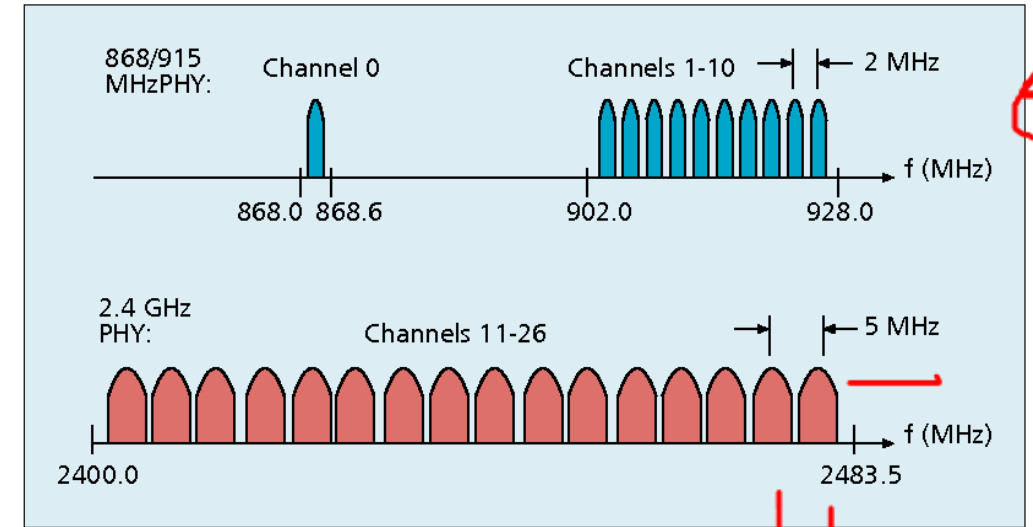
Show me the money: what is the actual data rate of 802.15.4 (2.4 GHz)?

- Chip rate: 2000 kchips/sec
- Occupies 2 MHz bandwidth



802.15.4 RF channels

- 27 channels across three bands
- 5 MHz channel separation at 2.4 GHz
 - Compare to 2 MHz for BLE
 - (or to 1 MHz for BT Classic)



Regional bands

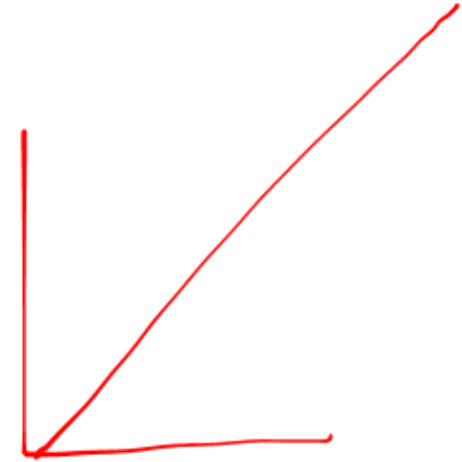
- Different RF bands have different regional availability
- Also have different rules
 - 915 MHz: 400 ms dwell time
 - 868 MHz: 1% duty cycle

	Channel	Center Frequency (MHz)	Availability
868 MHz Band	0	868.3	 <i>Europe</i>
	1	906	
915 MHz Band	2	908	 <i>Americas</i>
	3	910	
	4	912	
	5	914	
	6	916	
	7	918	
	8	920	
	9	922	
	10	924	
	11	2405	
2.4 GHz Band	12	2410	 <i>World Wide</i>
	13	2415	
	14	2420	
	15	2425	
	16	2430	
	17	2435	
	18	2440	
	19	2445	
	20	2450	
	21	2455	
	22	2460	
	23	2465	
	24	2470	
	25	2475	
	26	2480	

Signal strength

- Transmit power
 - Typical: 0 dBm (remember: 1 mW)

- Receiver sensitivity
 - nRF52840 802.15.4: -100 dBm
 - Compare to BLE sensitivity of -95 dBm
 - Minimum acceptable: -85 dBm
 - Circa-2006 radios (CC2420): -95 dBm
- Which has longer range, 802.15.4 or BLE? Why?
 - 802.15.4 with +5 dBm more margin; lower bit rate plays into this

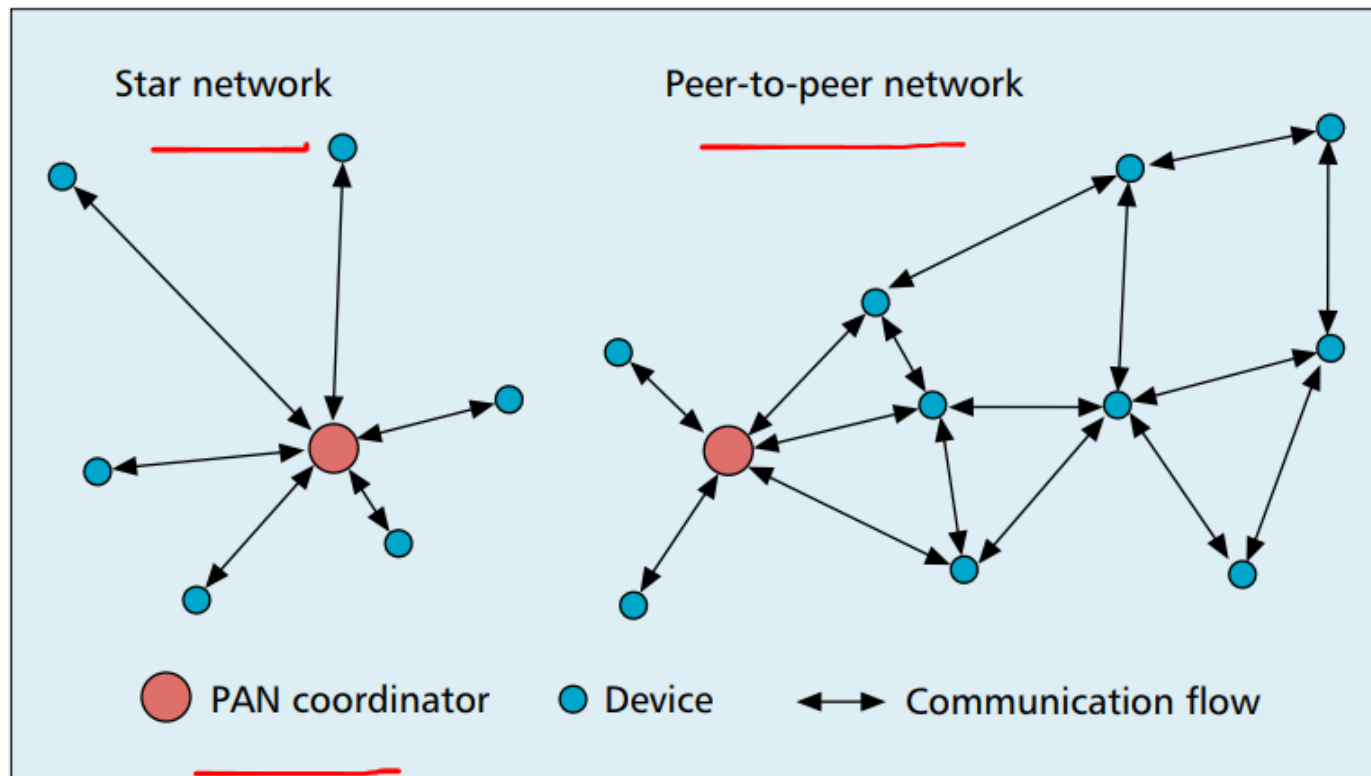


Outline

- Overview
- Physical Layer
- **Link Layer**

802.15.4 network topologies

- Only specifies PHY and MAC, but has use cases in mind



Star and Tree topologies

- PAN Coordinator
 - Receives and relays all messages
 - Most capable and power-intensive
 - Coordinators (a.k.a. Routers)
 - Control “clusters”
 - Receives and relays to its children
 - Communicates up to parent coordinator
 - End Devices
 - Only communicate with single parent coordinator
 - Least capable and power intensive
-

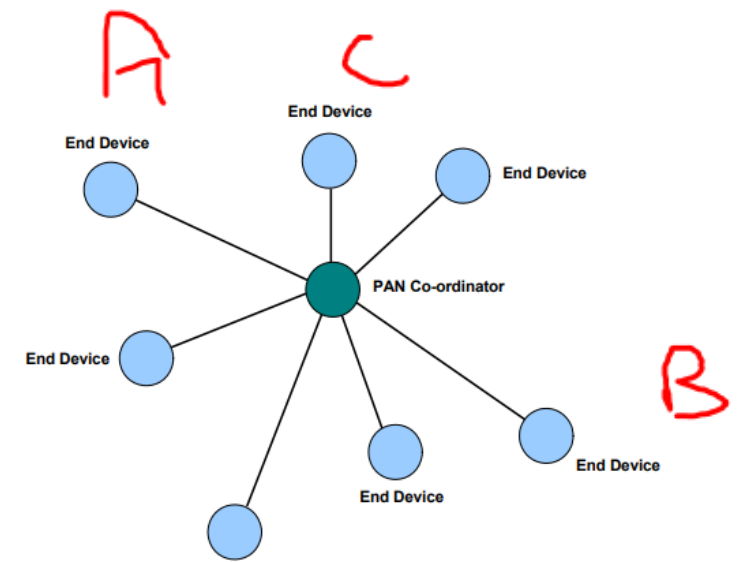


Figure 1: Star Topology

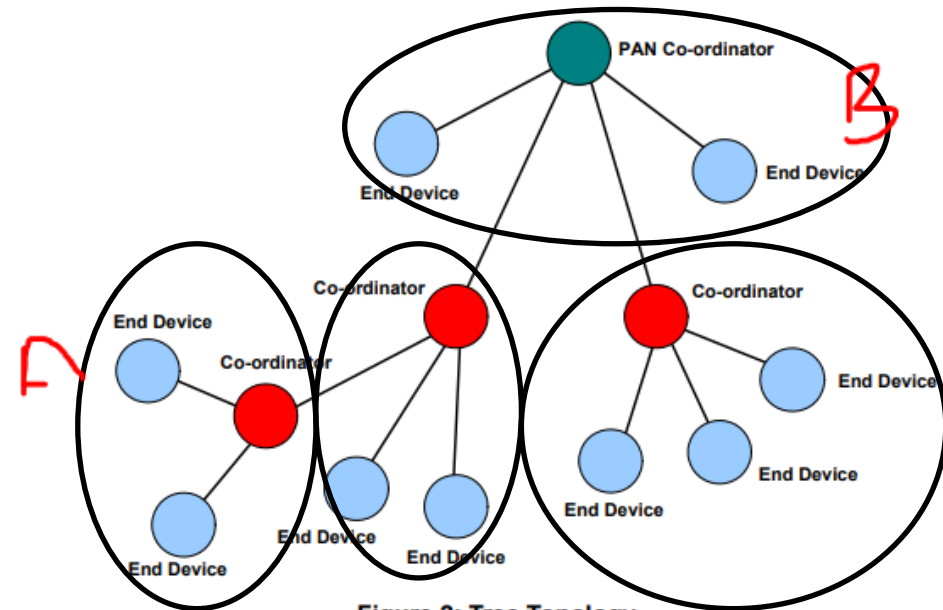


Figure 2: Tree Topology

Mesh networks

- Most devices are capable of communicating with multiple neighbors
- **What are advantages of mesh?**
- **What are disadvantages of mesh?**

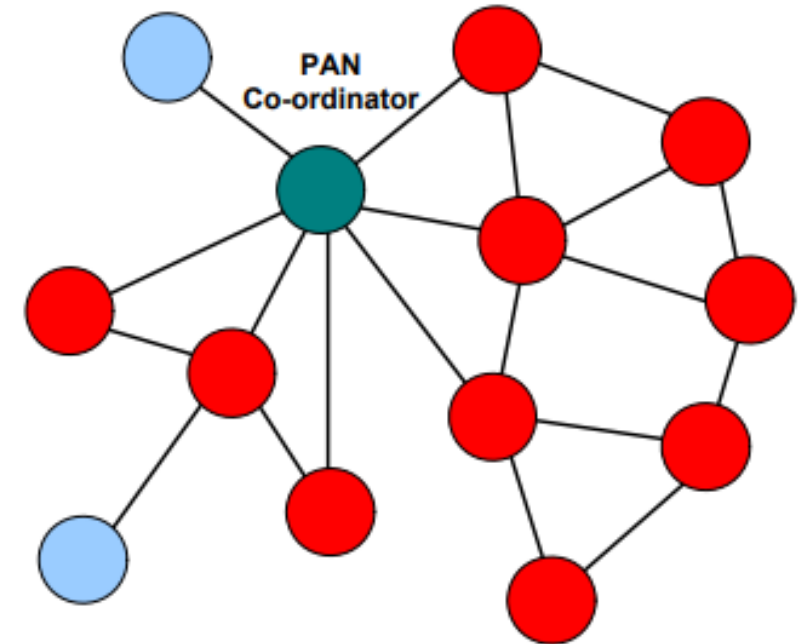


Figure 4: Mesh Topology



Mesh networks

- Most devices are capable of communicating with multiple neighbors
- **What are advantages of mesh?**
 - Devices can communicate over longer distances
 - Lower energy due to inverse d^2 path loss law
 - Device failures less likely to collapse the entire network
- **What are disadvantages of mesh?**
 - Some nodes have to spend more energy communicating
 - Network protocol becomes more complicated to manage routing

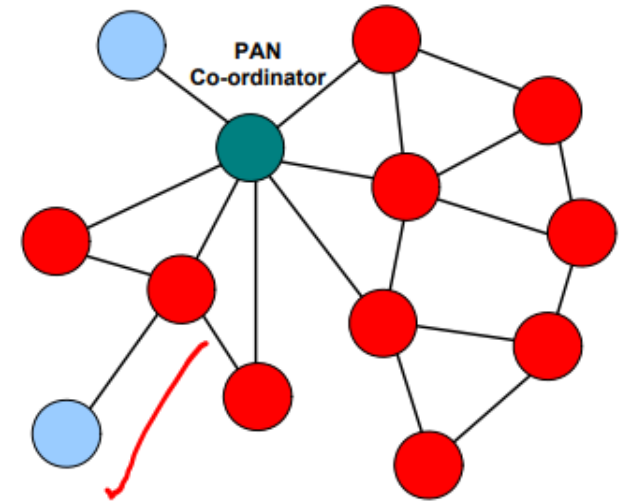


Figure 4: Mesh Topology

Reminder: CSMA/CA

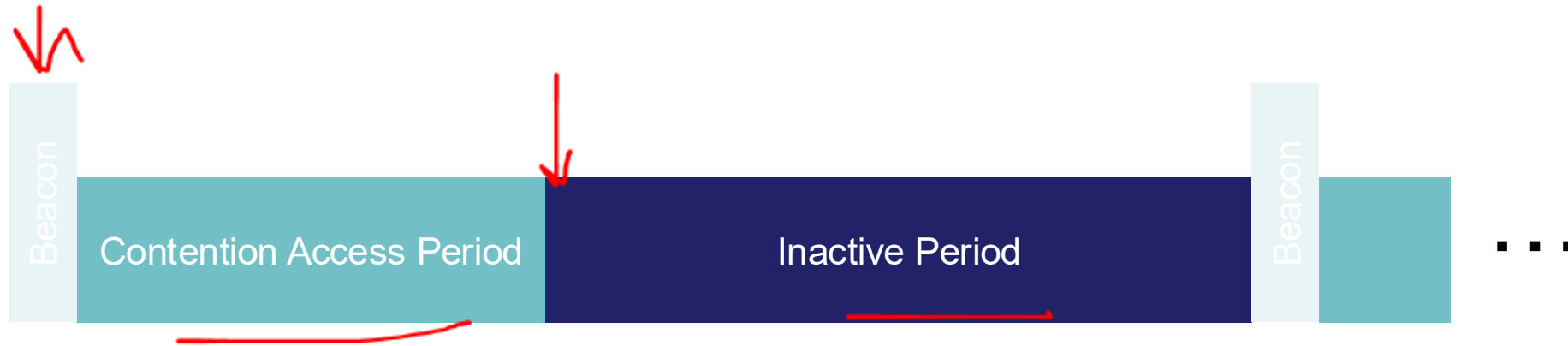
Carrier Sense Multiple Access with Collision Avoidance

1. First, wait a random amount (collision avoidance part)
 2. Then, listen for a duration and determine if anyone is transmitting (carrier sense part)
 - If idle, you can transmit
 - If busy, repeat step 1 (often increasing maximum wait time)
-
- Can be combined with notion of slotting
 - Synchronize to slots (smaller than transmit times)
 - Wait for a number of slots
 - Listen for idle slots

Modes of operation

- Beacon-enabled PAN
 - Slotted CSMA/CA
 - Structured communication patterns
 - Optionally with some TDMA scheduled slots
- Non-beacon-enabled PAN
 - Unslotted CSMA/CA
 - No particular structure for communication
 - Could be defined by other specifications, like Thread or Zigbee

Beacon-enabled superframe structure



- Beacons occur periodically [15 ms – 245 seconds]
 - Devices must listen to each beacon
- Contention Access Period
 - Slotted CSMA/CA synchronized by beacon start time
- Inactive Period
 - No communication occurring. Assumes sleepy devices

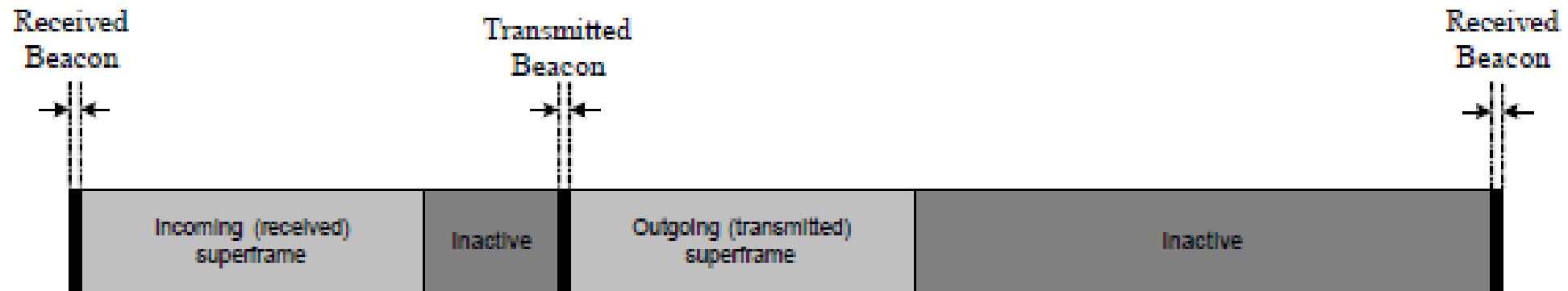
Guaranteed Time Slots (GTS)



- PAN Coordinator may create a Contention Free Period with Guaranteed Time Slots
 - TDMA schedule assigned to specific devices
 - Slots eat up part of the Contention Access Period
 - No CSMA/CA within a slot

Handling tree-based topologies

- All coordinators listen to beacon from PAN coordinator
 - And can participate in that contention period
- Send their own beacons to leaf devices during inactive period
 - Leaves participate in that contention period



Receiving messages

1. Listen during entire contention period
 - Can receive direct messages from any other device
 - Can immediately respond to messages as well
2. Request messages from Coordinator
 - Make all communication go through Coordinator
 - Send a request-for-data packet to coordinator to get information
 - Coordinator can include list of devices with pending data in beacon
- More complicated listening algorithms are possible

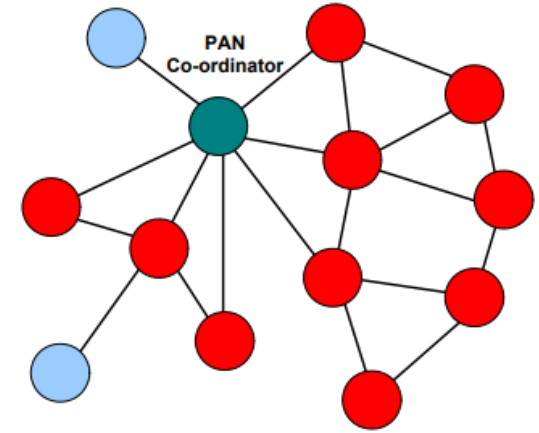


Figure 4: Mesh Topology

Question

- What are benefits/costs of using or not using beacons?

Question

- What are benefits/costs of using or not using beacons?
 - Beacons
 - Enable energy savings by designating period with radios off
 - Enable structured communication like Guaranteed Slots
 - Require some central coordinator within range of all devices
 - Tradeoff in inactive period:
 - communication latency vs beacon-listening costs
 - No beacons
 - Enable all devices to be identical (no coordinator needed)
 - Require / permit custom communication scheme